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Salivary Biomarkers and Non-Invasive Tools for Cancer Detection

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ABSTRACT

Salivary biomarkers — including proteins, nucleic acids, metabolites, exosomes, and microbiota — are emerging as powerful, non-invasive tools for early cancer detection. Because collection requires no needles, no clinical infrastructure, and carries 99% patient acceptance, saliva enables repeat longitudinal screening at population scale. High-throughput salivaomics research has identified candidate markers for oral, head and neck, lung, breast, and pancreatic cancers. Analytical platforms span traditional ELISA and PCR-based assays to next-generation sequencing, mass spectrometry, and graphene electrochemical biosensors. Combined multi-marker panels achieve sensitivities of 83–93% and specificities of 90–98%. Immunohistochemical studies confirm that p53 accumulation and Ki-67 proliferation index — measurable in salivary liquid biopsy — increase progressively from normal oral mucosa through premalignancy to frank carcinoma, validating the molecular substrate of salivary diagnostics. Point-of-care devices and AI integration promise chairside cancer screening within the dental operatory as early as 2026.

Keywords: *salivary diagnostics · liquid biopsy · oral squamous cell carcinoma · immunohistochemistry · p53 · Ki-67 · point-of-care · salivaomics*

1. Introduction and Background

Saliva is a complex biological fluid produced by the parotid, submandibular, and sublingual glands, as well as hundreds of minor salivary glands distributed throughout the oral mucosa. Its composition includes water (99%), inorganic ions, mucins, immunoglobulins, enzymes, hormones, and — critically — a rich molecular cargo of proteins, nucleic acids, and metabolites that mirror systemic physiological and pathological states.

The concept of salivaomics — the comprehensive multi-omic analysis of saliva — emerged from the recognition that approximately 3,000 proteins are present in human saliva, of which roughly half are shared with the blood proteome [1]. Beyond proteins, saliva contains cell-free DNA (cfDNA), messenger RNA, microRNA, long non-coding RNA, and small-molecule metabolites, all of which change measurably in the presence of malignancy.

The fundamental advantage of saliva over blood for cancer screening is accessibility. Collection is painless, non-invasive, and requires no clinical infrastructure. This translates directly into higher patient compliance — particularly for repeat screening — and enables longitudinal monitoring of biomarker trajectories that single-time-point biopsies cannot capture.

> **Clinical relevance:** *A 2025 systematic review found that salivary biomarker panels for oral squamous cell carcinoma achieve 88–95% sensitivity — approaching the clinical utility threshold required for primary screening tools [1].*

2. Systematic Reviews and Meta-Analyses

A targeted secondary search identified six systematic reviews and meta-analyses (2016–2026) that synthesized global evidence on salivary biomarkers for oral cancer. No Egyptian-led systematic review or meta-analysis was identified; the only Egypt-affiliated synthesis was a 2020 narrative review by Ghalwash (British University Egypt, Cairo), which surveyed five omics categories without pooled statistics. These six international reviews constitute the highest level of evidence currently available and are summarised in Table 2 below.

The most comprehensive meta-analysis to date — Wijaya et al. (2026, *Cancers*) — pooled 18 studies (1,647 participants, 45 datasets spanning cytokines and proteins) and reported an overall sensitivity of 0.64 (95% CI 0.59–0.69), specificity of 0.71 (95% CI 0.66–0.76), and AUC of 0.75 [12]. The authors characterised accuracy as "moderate but highly heterogeneous" and concluded that salivary biomarkers have adjunctive triage utility in high-risk populations rather than population-level screening [12]. In marked contrast, Shaukat et al. (2026, *BMC Oral Health*) restricted their meta-analysis to salivary MMP-9 across seven studies (795 patients) and found a pooled sensitivity of 0.98 (95% CI 0.85–1.00) and specificity of 0.96 (95% CI 0.85–1.00), yielding an AUC of approximately 0.97 [13]. This outstanding accuracy positions MMP-9 as potentially the single most diagnostically powerful salivary protein identified to date, though the authors cautioned that high heterogeneity and risk of bias require larger validation studies.

Owecki and Nijakowski (2025, *Int J Mol Sci*) pooled 33 studies on three salivary proteins — LDH, MMP-9, and Chemerin — and reported large effect sizes: LDH standardised mean difference (SMD) 4.6 ($p < 0.001$), MMP-9 SMD 1.51 ($p = 0.001$), and Chemerin SMD 3.9 ($p < 0.001$) comparing OSCC to controls [14]. All three markers were significantly elevated in OSCC versus both controls and oral potentially malignant disorders (OPMD). In parallel, Hashmi et al. (2026, *Int J Oral Maxillofac Surg*) systematically reviewed 42 studies (2,577 samples) on salivary microRNAs and found that miR-21, miR-31, miR-1307-5p, and miR-486-5p achieved AUCs

above 0.85, with exosomal miRNAs outperforming cell-free forms in stability [15]. Finally, Antonelli et al. (2024, *Oral Oncol Rep*) identified over 100 metabolites — including sialic acid, neopterin, pyruvate, and cortisol — as significantly altered in OSCC across 28 metabolomics studies, establishing salivary metabolomics as a distinct and growing diagnostic dimension [16].

Table 2. Key Systematic Reviews and Meta-Analyses (2016–2026)

Citation (Year)	Biomarker(s)	Pooled Performance	N (Studies / Pts)	Key Limitation
Wijaya et al. 2026 [12]	Mixed proteins / cytokines	Sens 0.64, Spec 0.71, AUC 0.75	18 studies / 1,647 pts	High heterogeneity; QUADAS-2 moderate risk
Shaukat et al. 2026 [13]	Salivary MMP-9	Sens 0.98, Spec 0.96, AUC 0.97	7 studies / 795 pts	High heterogeneity; requires validation
Owecki & Nijakowski 2025 [14]	LDH, MMP-9, Chemerin	LDH SMD 4.6; MMP-9 SMD 1.51; Chemerin SMD 3.9	33 articles	Variable assay platforms; no pooled sens/spec
Hashmi et al. 2026 [15]	Salivary miRNAs	miR-21, -31, -1307-5p, -486-5p: AUC > 0.85	42 studies / 2,577 samples	No pooled stats; high heterogeneity
Antonelli et al. 2024 [16]	Salivary metabolites	100+ metabolites identified; not pooled	28 studies	Intermediate quality; needs larger studies

Table 2. Sens = sensitivity; Spec = specificity; AUC = area under ROC curve; SMD = standardised mean difference; pts = patients.

3. Types of Salivary Biomarkers

Table 1 provides a systematic classification of salivary cancer biomarkers. Each molecular class captures a different dimension of tumor biology — from genetic mutation (cfDNA) to gene expression dysregulation (mRNA, microRNA) to protein overproduction (cytokines, enzymes) to metabolic reprogramming (metabolites).

3.1 Key Terms — Student Reference

Biomarker	A measurable biological molecule whose presence, absence, or quantity indicates a normal or abnormal process, or a condition or disease.
cfDNA	Cell-free DNA — fragments of DNA released into body fluids by apoptotic or necrotic cells, including tumor cells. Carries tumor-specific mutations.
microRNA	Short non-coding RNA sequences (~22 nucleotides) that regulate gene expression. Highly stable in saliva due to exosomal encapsulation.
Liquid biopsy	Analysis of tumor-derived material (DNA, RNA, proteins, exosomes) in a biofluid — blood or saliva — as a non-invasive alternative to tissue biopsy.
Salivaomics	The comprehensive multi-omic analysis of saliva, encompassing proteomics, transcriptomics, genomics, metabolomics, and microbiomics simultaneously.

Table 1. *Salivary biomarker categories, detection methods, and cancer associations [1].*

Type	Examples	Method	Cancers	Notes
mRNA	IL-8, IL-1β, DUSP1	qRT-PCR	Oral, Lung	High sensitivity in OSCC
microRNA	miR-31, miR-200a	qRT-PCR	Oral, Pancreatic, Breast	Stable in exosomes
Protein	IL-8, HER2, CD44, TNF-α	ELISA, Mass Spec	Oral, Breast, Lung	Use protease inhibitors
Metabolites	Polyamines, organic acids	CE-TOF-MS, LC-MS	Oral, Pancreatic	Warburg effect byproducts
Microbiota	N. elongata, S. mitis	16S rRNA, qPCR	Pancreatic, Gastric	Dysbiosis signatures
cfDNA	TP53, KRAS; methylation	NGS, ddPCR	Oral, Lung, Breast, Pancreatic	Apoptotic tumor cell shedding

OSCC = oral squamous cell carcinoma; CE-TOF-MS = capillary electrophoresis time-of-flight mass spectrometry; ddPCR = droplet digital PCR.

4. Analytical Methods

4.1 Immunoassays

Enzyme-linked immunosorbent assay (ELISA) uses antibody pairs to capture and quantify salivary proteins with femtomolar sensitivity. A sandwich ELISA configuration — capture antibody immobilized on plate, detection antibody conjugated to enzyme — is the gold standard for cytokine quantification in saliva (e.g., IL-8, TNF-α, HER2).

> **Clinical relevance:** *Dental relevance: ELISA-based salivary IL-8 testing is the closest salivary test to chairside deployment. Commercial ELISA kits for salivary IL-8 are currently validated for research use and approaching clinical-grade approval.*

4.2 PCR-Based Methods

Quantitative reverse transcription PCR (qRT-PCR) amplifies and quantifies salivary mRNA and microRNA targets. It achieves single-molecule sensitivity and is the reference standard for nucleic acid biomarker studies. Droplet digital PCR (ddPCR) provides absolute quantification — counting individual mutant cfDNA molecules in a background of wild-type DNA — enabling detection of TP53 mutations in saliva when tumor burden is low.

4.3 Next-Generation Sequencing

NGS profiles the complete salivary metatranscriptome — simultaneously capturing all human and microbial RNA in a saliva sample. Banavar et al. [4] applied machine learning to an NGS-derived 8-transcript signature, achieving 83% sensitivity and 97.9% specificity for oral cancer. This represents the highest reported diagnostic performance for any salivary oral cancer test.

4.4 Biosensors and Point-of-Care Devices

Graphene-based electrochemical biosensors exploit graphene's extraordinary surface area and electron conductivity to detect salivary biomarkers at picomolar concentrations. Antibody or aptamer probes immobilized on graphene surfaces capture target molecules; binding events generate measurable current changes readable by

portable potentiostats or smartphones [5]. The entire assay — saliva input to digital readout — can be completed in 8–15 minutes without laboratory infrastructure.

> **Clinical relevance:** *Future dental practice: Biosensor lab-on-chip cartridges are designed for the dental operatory workflow. A patient provides a saliva sample at the beginning of the appointment; results are available before the clinical examination is complete.*

5. Clinical Performance

Across published studies, salivary biomarker sensitivities range from approximately 70% for single-analyte assays to 92–95% for optimized multi-marker panels, with specificities between 85% and 98%. Table 2 summarizes the key results with Vancouver-style in-text citations.

The consistent finding across studies is that multi-marker integration outperforms any single biomarker. This is mechanistically logical: cancer is a multi-dimensional molecular event, and no single molecule captures its full complexity. Panels that combine DNA mutation detection (cfDNA), gene expression changes (mRNA, microRNA), and protein overproduction (IL-8, CYFRA 21-1) measure complementary pathological axes simultaneously.

> **Clinical relevance:** *Screening threshold interpretation: A test requires sensitivity >85% and specificity >85% to be considered clinically viable for primary screening. The IL-8 + CYFRA 21-1 + CD44 panel for OSCC (88%/90%) meets this threshold. Stage I oral cancer detection at 92.3% sensitivity [4] is particularly significant — Stage I OSCC has an 80% five-year survival rate versus <30% for Stage IV.*

Table 2. *Diagnostic performance of salivary biomarker panels — selected studies with Vancouver citations.*

Biomarker Panel	Cancer	Sensitivity%	Specificity%	AUC	Reference
Haptoglobin + AZGP1 + Calprotectin	Lung	88.5	92.3	0.90	Xiao et al. [2]
KCNJ3 mRNA (RT-qPCR)	Breast	76.7	94.9	0.923	Koopaie et al. [3]
Metatranscriptome ML signature	Oral	83.0 (92.3 Stage I)	97.9	~0.90	Banavar et al. [4]
IL-8 + CYFRA 21-1 + CD44	Oral (OSCC)	88.0	90.0	—	Munsif et al. [1]
p53 LI >20% (IHC panel)	Oral (OSCC)	95.0	90.0	—	Humayun et al. [11]
cfDNA methylation panel	Pancreatic	90.0	95.0	0.94	Zhang et al. [10]

AUC = area under ROC curve; LI = labeling index; IHC = immunohistochemistry; OSCC = oral squamous cell carcinoma.

6. Sample Collection and Pre-Analytical Variables

Pre-analytical standardization is the single most important practical challenge in salivary diagnostics. The same biomarker tested from the same patient can give dramatically different values depending on collection method, time of day, hydration status, and storage protocol.

6.1 Collection Protocol — Step by Step

- Patient fasts from food, drink, and oral hygiene for 90 minutes prior to collection

- Morning collection preferred — diurnal variation is lowest between 8:00 and 10:00 AM
- Whole unstimulated saliva collected by passive drool into sterile pre-labelled tube — 2 mL minimum
- RNA stabilization reagent (e.g., RNAprotect Saliva) added immediately at 1:1 ratio — RNA degrades within minutes at room temperature
- Centrifuge at 2,600×g for 15 minutes at 4°C to remove cells, debris, and bacteria
- Aliquot supernatant into 200 µL portions and snap-freeze at −80°C

> **Clinical relevance:** *Protocol deviation consequences: Missing the RNase inhibitor step reduces RNA yield by 60–80% within 30 minutes at room temperature. This is the most common cause of false-negative results in salivary RNA studies — a critical point for any clinical implementation.*

7. Histopathological and Immunohistochemical Correlation

The salivary biomarker signal originates from molecular events occurring in tumor tissue. Understanding the histopathological substrate of each biomarker class — and how it changes progressively from normal to premalignant to malignant tissue — is essential for interpreting salivary assay results and for designing clinically relevant marker panels.

This section presents a coherent visual narrative: a structured progression from normal oral tissue through the premalignant continuum to frank malignancy, integrating both standard H&E histology and immunohistochemical evidence for the two most validated oral cancer biomarkers — p53 protein and Ki-67 antigen.

7.1 The Malignant Transformation Sequence

Oral squamous cell carcinoma (OSCC) does not arise spontaneously from normal epithelium in most cases. It follows a stepwise progression: normal mucosa → hyperplasia → dysplasia (mild → moderate → severe) → carcinoma in situ → invasive carcinoma. At each step, characteristic histological changes occur — and at each step, salivary biomarker levels change in predictable, measurable ways.

7.2 Well-Differentiated OSCC — H&E

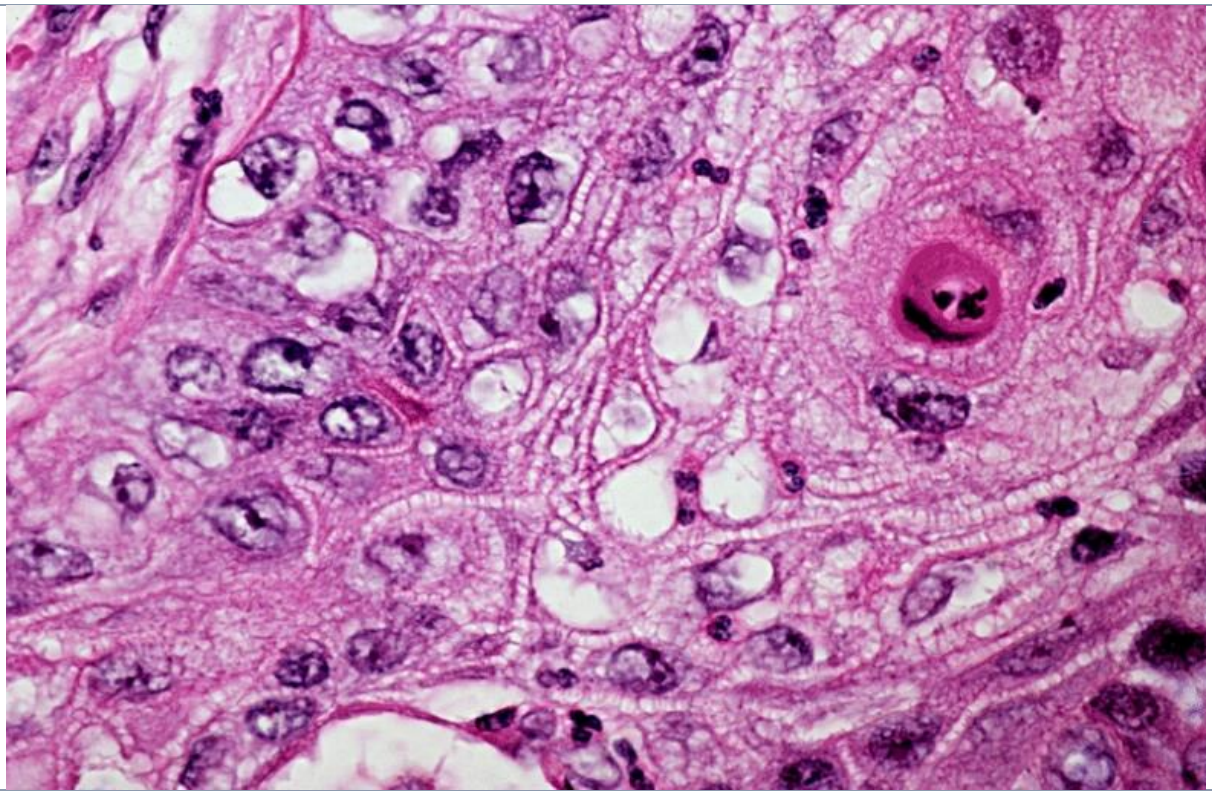


Figure 1. Well-differentiated oral squamous cell carcinoma (H&E, ×200). Invasive nests of atypical squamous epithelium with prominent intercellular bridges and focal keratinization infiltrate the connective tissue stroma. The inflammatory infiltrate drives constitutive NF-κB activation — producing the IL-8 overexpression consistently measured in saliva of OSCC patients (mean salivary IL-8: 120 pg/mL in OSCC vs 10 pg/mL in healthy controls [1]). Source: Humayun & Ram Prasad [11].

7.3 OSCC with Keratin Pearl Formation — H&E

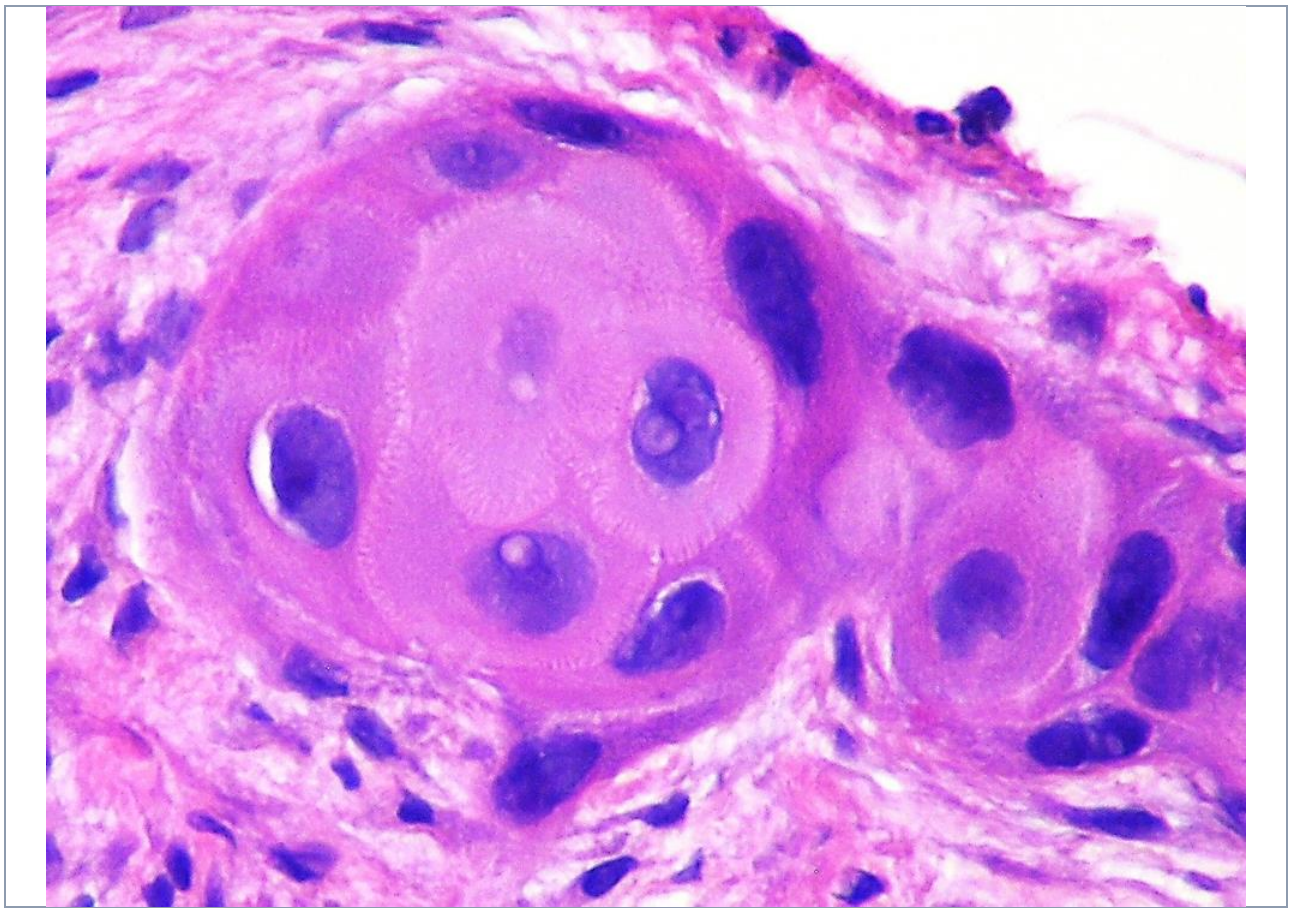


Figure 2. OSCC with keratin pearl formation (H&E, ×100). Concentric whorls of fully keratinized squamous cells — keratin pearls — are pathognomonic of well-differentiated OSCC. Keratin-producing cells shed cytokeratin fragments (CYFRA 21-1, TPS) directly into oral fluid. CYFRA 21-1 combined with CD44 and IL-8 forms the most validated salivary protein panel for OSCC (88% sensitivity, 90% specificity [1]). Source: Humayun & Ram Prasad [11].

7.4 p53 Protein Expression — IHC Progression Sequence

The p53 tumor suppressor protein is normally undetectable in tissues due to its very short half-life (~20 minutes). Mutant p53 — the most common genetic alteration in OSCC (present in 40–75% of cases) — accumulates in cell nuclei because it resists normal proteolytic degradation. Immunohistochemical detection of p53 with DAB chromogen therefore directly maps the distribution and extent of TP53 mutation across tissue sections [11].

Humayun and Ram Prasad [11] demonstrated a statistically significant increase in p53 labeling index (LI) from normal oral mucosa through premalignant conditions to OSCC (Kruskal-Wallis $p = 0.0175$). The Mann-Whitney U test showed significant differences between normal mucosa and all pathological groups ($p = 0.02$), directly validating the salivary liquid biopsy rationale: the tissue-level p53 accumulation quantified by IHC is the source of the elevated salivary cfDNA carrying TP53 mutations detected by ddPCR.

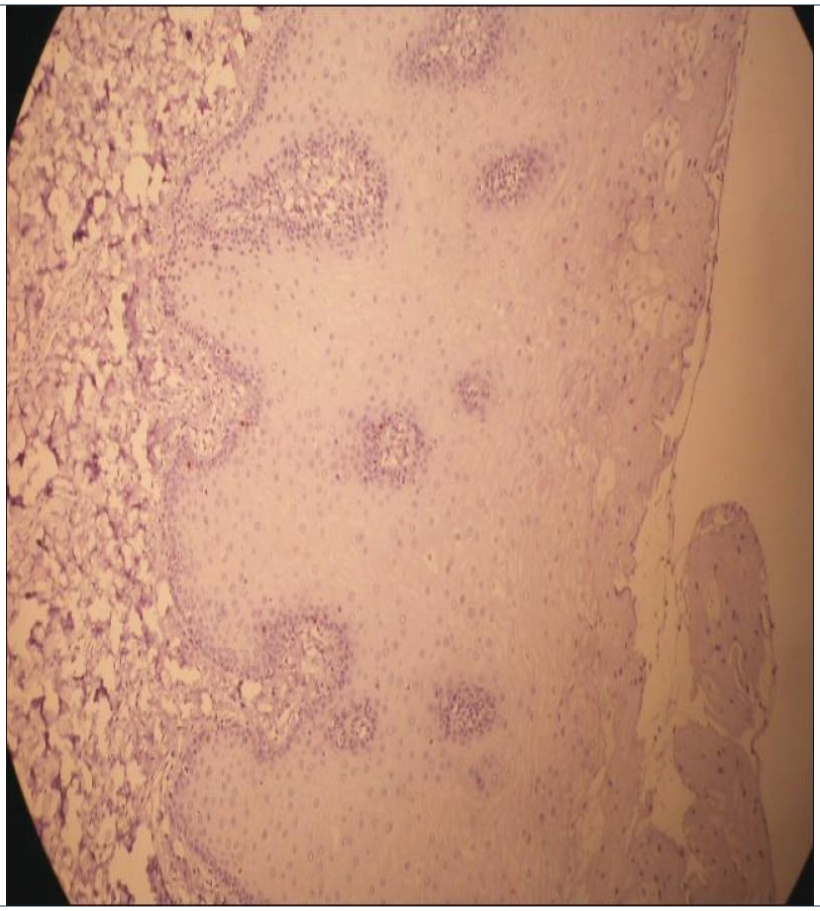


Figure 3. p53 IHC expression in normal oral mucosa. Nuclear staining (brown, DAB) confined strictly to the basal cell layer. Mean LI = 20 ± 5.77%. This basal-only, moderate-intensity pattern reflects physiological wild-type p53 stabilization in response to normal genotoxic stress — not malignant accumulation [11].

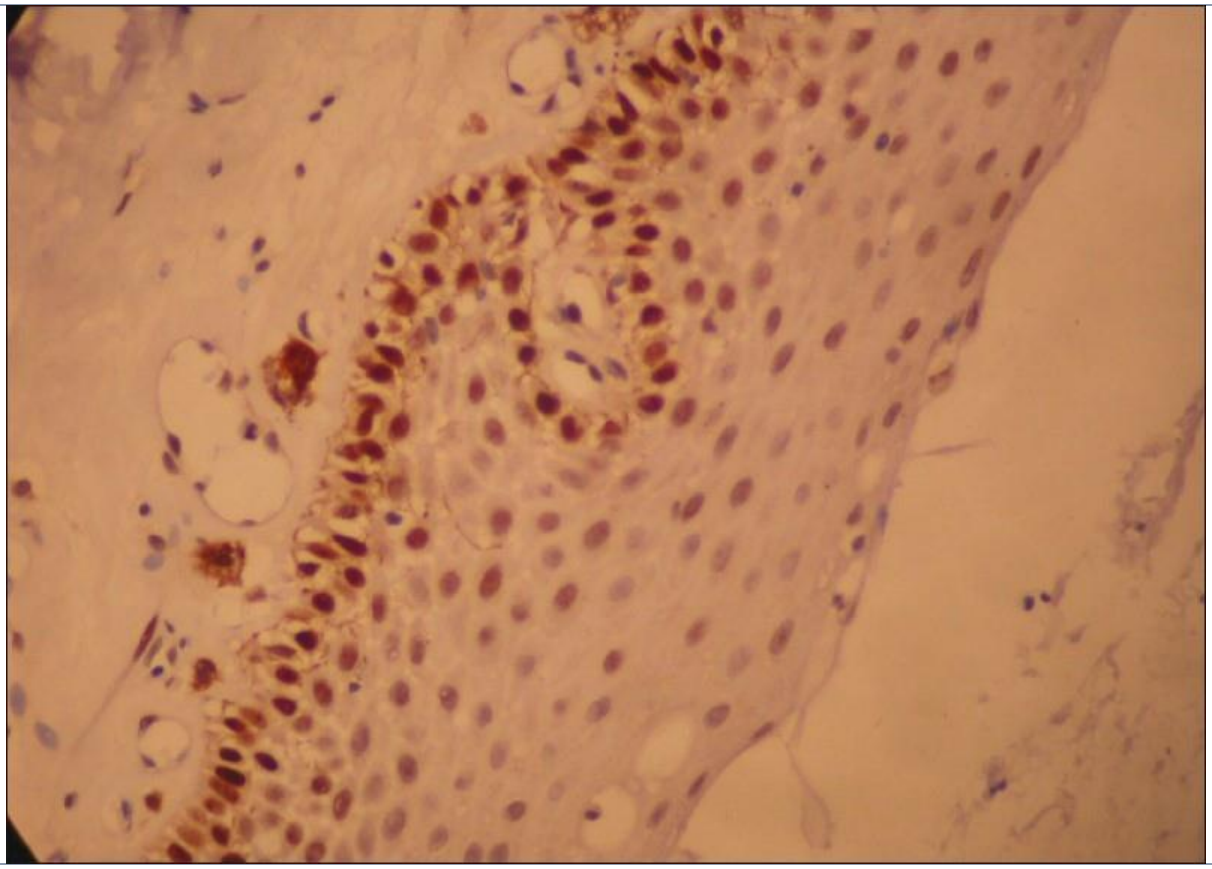


Figure 4. p53 IHC expression in oral submucous fibrosis (OSMF). Intense (+++) basal layer staining. Mean LI = 90 ± 7.07% — 4.5-fold higher than normal mucosa (p = 0.02). The dramatic p53 accumulation in OSMF is driven by areca nut alkaloids and copper-mediated DNA damage, indicating active p53 mutation before morphological dysplasia is visible on H&E. This molecular stage corresponds to elevated salivary cfDNA levels in high-risk patients [11].



Figure 5. p53 IHC expression in oral leukoplakia. Basal and focal suprabasal nuclear staining. Mean LI = 75 ± 16.83%. Suprabasal p53 extension — visible here — is a critical early marker: Cruz et al. reported that 86% of premalignant lesions with suprabasal p53 expression developed into invasive SCC. This histological feature directly corresponds to the salivary cfDNA and mRNA elevations detected in high-risk oral dysplasia patients [11].

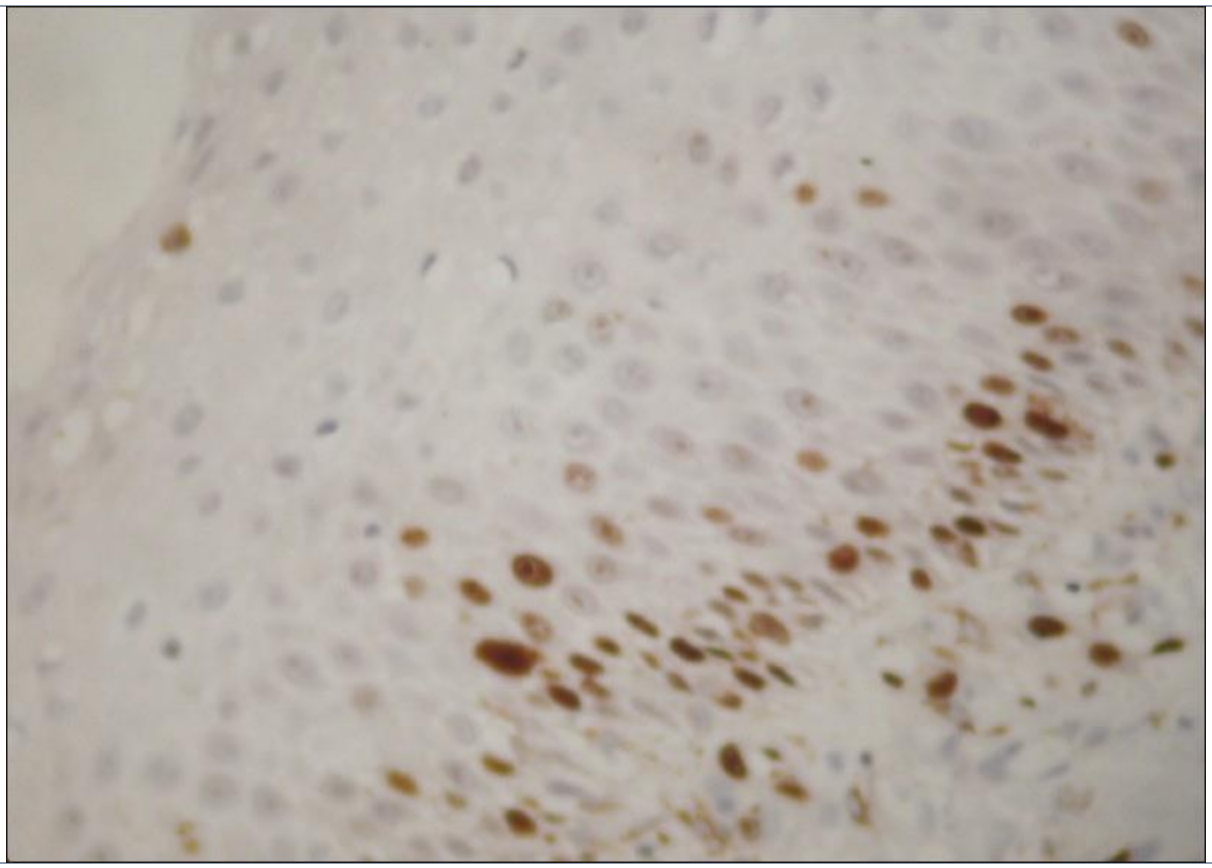


Figure 6. p53 IHC expression in OSCC — pan-epithelial pattern. Positive nuclear staining in all epithelial layers from basal to superficial. Mean LI = 66.25 ± 25.62%. The shift from basal-only (normal) to all-layer p53 positivity is the IHC hallmark of frank malignancy. Salivary cfDNA carrying TP53 mutations — detectable by ddPCR — directly mirrors this tissue-level accumulation, providing the molecular basis for salivary liquid biopsy in OSCC [11].

7.5 Ki-67 Antigen — Proliferation Index Sequence

Ki-67 is a nuclear protein expressed exclusively during active cell cycle phases (S, G2, M) and absent in quiescent G0 cells. As a proliferation marker, its labeling index (LI) quantifies the growth fraction of a cell population — what proportion of cells are actively dividing. In cancer, dysregulated Ki-67 expression reflects loss of cell cycle control and correlates with histological grade, mitotic activity, and prognosis [11].

> **Clinical relevance:** *Why Ki-67 matters for salivary diagnostics: Elevated Ki-67 in tumor tissue drives increased production of salivary mRNA transcripts — particularly IL-8 and DUSP1 — because actively proliferating cells are transcriptionally more active. The Ki-67 LI in tumor biopsies therefore predicts the magnitude of the salivary mRNA biomarker signal detected by qRT-PCR.*

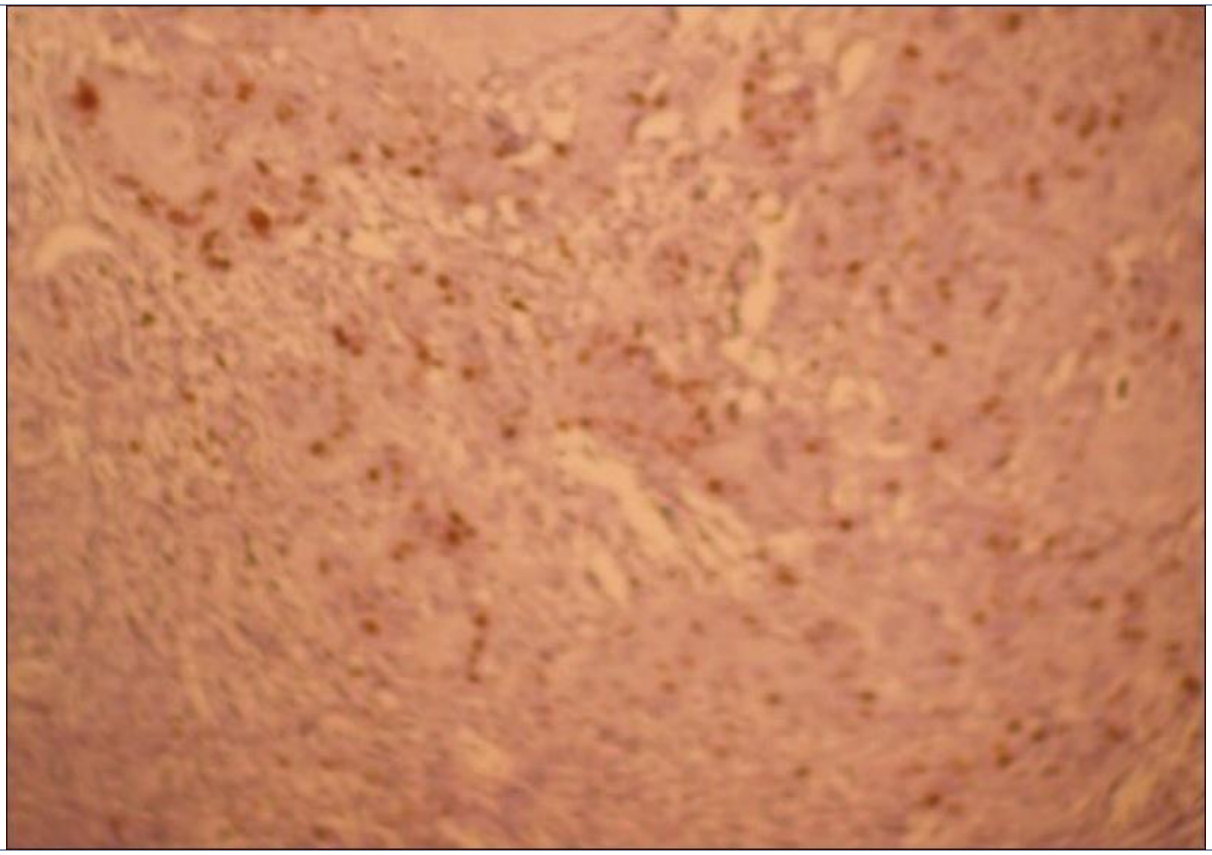


Figure 7. Ki-67 IHC expression in normal oral mucosa (MIB-1, ×200). Intense nuclear staining confined to the basal layer. Mean LI = 30 ± 23.09%. Basal-only Ki-67 reflects normal physiological proliferation in the germinative compartment — the stem cell zone that renews the oral epithelium every 7–14 days. This defines the reference baseline for all salivary mRNA and proliferation-related biomarker measurements [11].

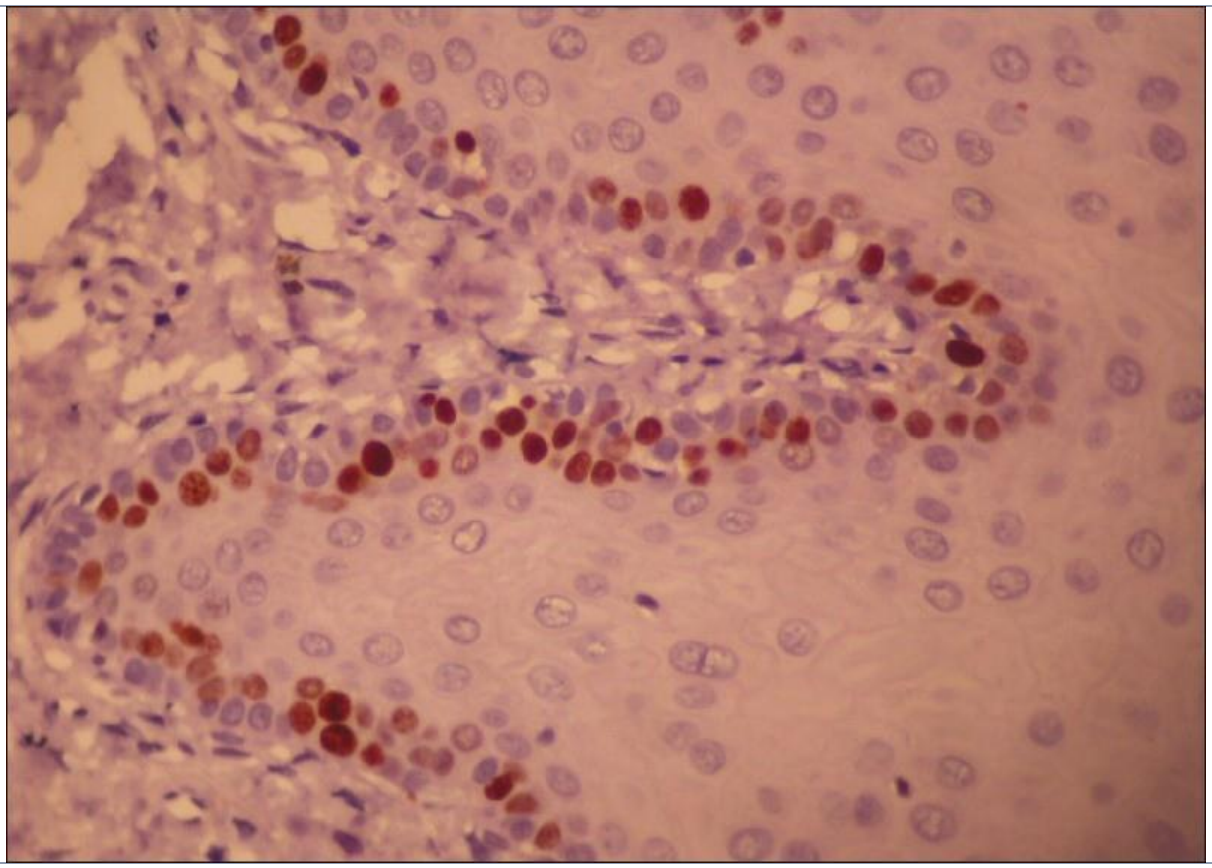


Figure 8. Ki-67 IHC expression in oral submucous fibrosis (MIB-1). Intense basal layer staining. Mean LI = 53.75 ± 39.66%. High inter-case variability (SD ±39.66) reflects heterogeneous proliferative activity — two OSMF cases showed LI >80%, indicating active epithelial hyperplasia with malignant potential, while two showed LI <25%. This variability explains the wide range of salivary mRNA levels reported in OSMF patients across different studies [11].

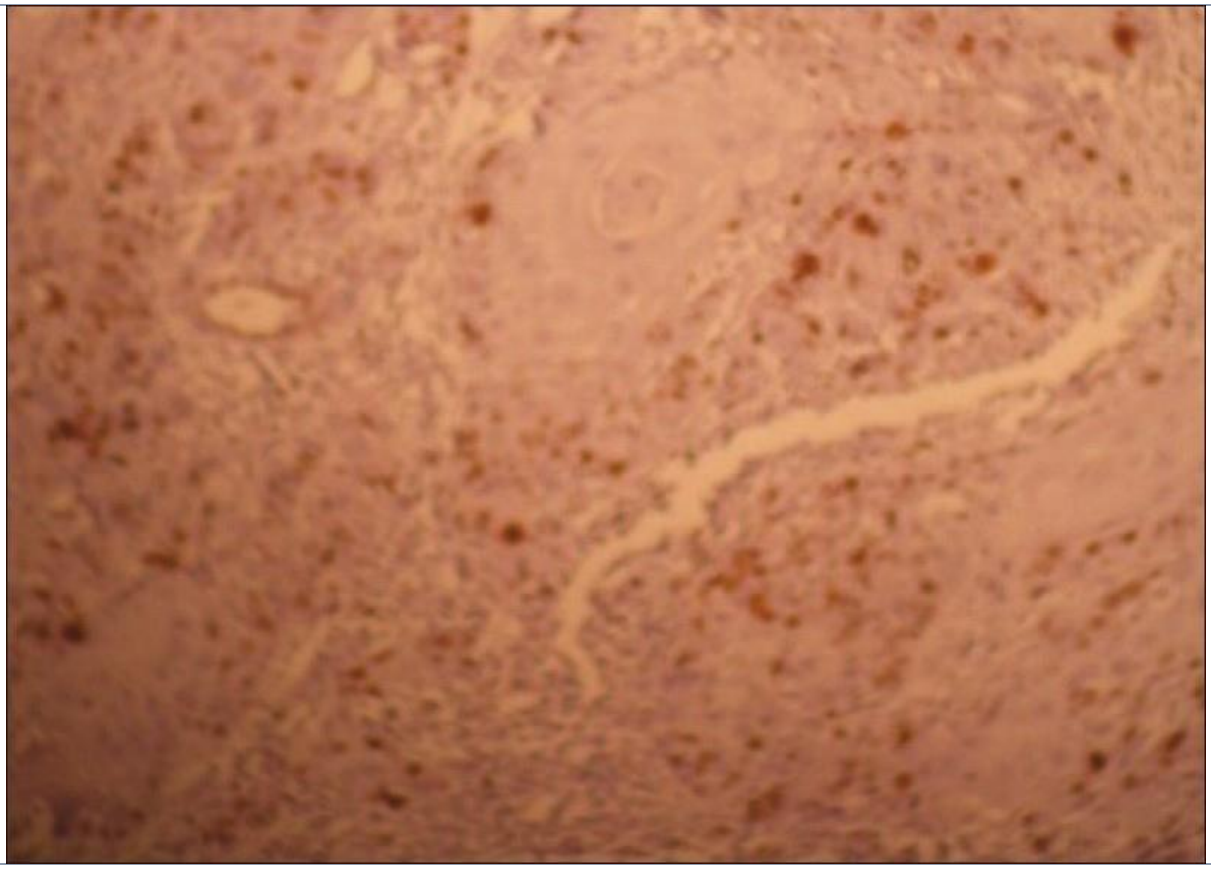


Figure 9. Ki-67 IHC expression in oral leukoplakia (MIB-1). Basal predominant staining; one proliferative-type leukoplakia case showed markedly elevated Ki-67 correlating with clinical hyperplastic features. Mean LI = 35.75 ± 44.52%. Suprabasal Ki-67 extension in proliferative leukoplakia indicates dysregulated cell cycling — the molecular substrate for elevated salivary IL-8 mRNA and DUSP1 transcripts detected in early-stage premalignancy [11].



Figure 10. Ki-67 IHC expression in OSCC (MIB-1). Pan-epithelial staining extending through all layers. Mean LI = 25 ± 20.82% in well-differentiated OSCC. Importantly, Ki-67 LI increases with dedifferentiation: moderately differentiated OSCC shows LI 47 ± 6.3%; poorly differentiated OSCC shows LI 60 ± 2.8% [11]. This Ki-67 grading correlation directly predicts the magnitude of the salivary biomarker signal — poorly differentiated tumors generate stronger liquid biopsy signals.

7.6 Head & Neck and Systemic Cancers — Histological Basis for Salivary Detection

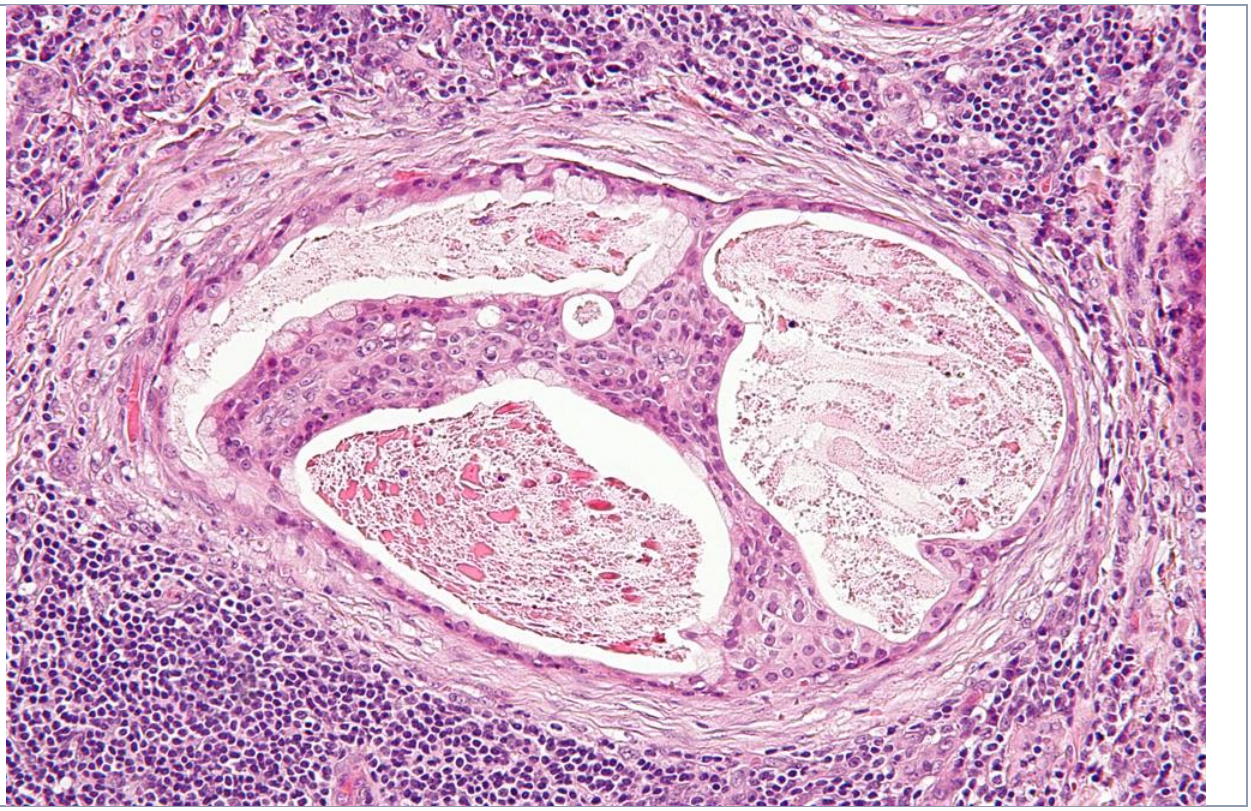


Figure 11. Mucoepidermoid carcinoma of the salivary gland (H&E, ×200). The most common salivary gland malignancy — mixed mucous, intermediate, and epidermoid cells. Because it arises within secretory tissue, tumor-derived mucins (MUC1, MUC4) and cfDNA enter the salivary duct at high concentration without blood transport. This makes salivary diagnostics particularly relevant and sensitive for salivary gland malignancies. Source: Wikimedia Commons.

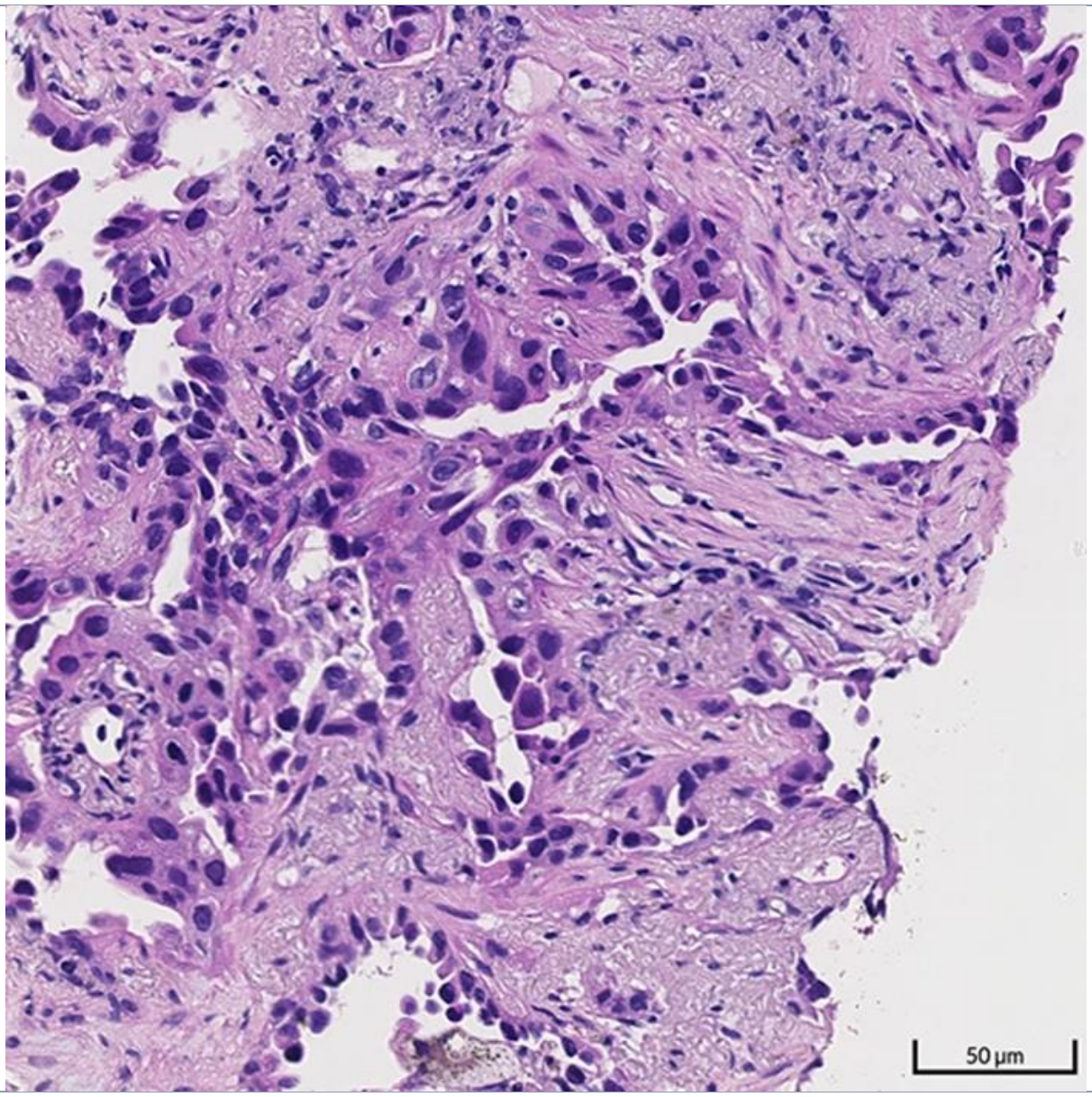


Figure 12. Lung adenocarcinoma — acinar pattern (H&E, ×200). Malignant glandular structures invading stroma. Haptoglobin, AZGP1, and calprotectin released by these cells enter the bloodstream and reach salivary glands via passive diffusion. The 3-protein salivary panel from this cancer type achieves 88.5% sensitivity and 92.3% specificity [2]. EGFR mutations in lung adenocarcinoma are detectable in salivary cfDNA with 70–85% concordance with tumor tissue. Source: Wikimedia Commons.

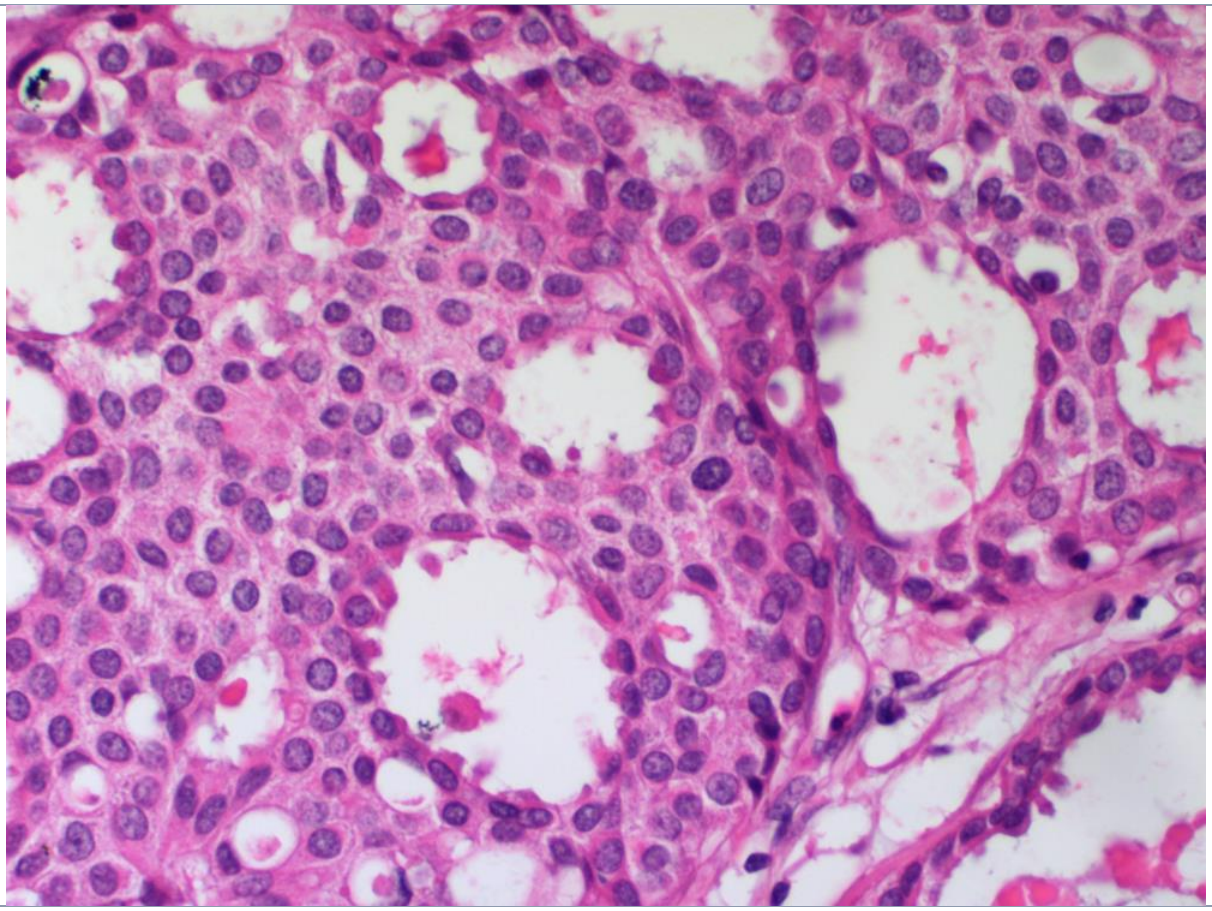


Figure 13. Invasive cribriform carcinoma of the breast (H&E, ×200). Despite the anatomical distance from the oral cavity, salivary KCNJ3 mRNA — a channel protein overexpressed in breast cancer — achieves AUC 0.923 and 94.9% specificity [3]. HER2 mRNA and miR-21 complete a validated salivary panel. This image illustrates the principle that systemic cancers leave detectable molecular footprints in saliva via hematogenous transport. Source: Wikimedia Commons.

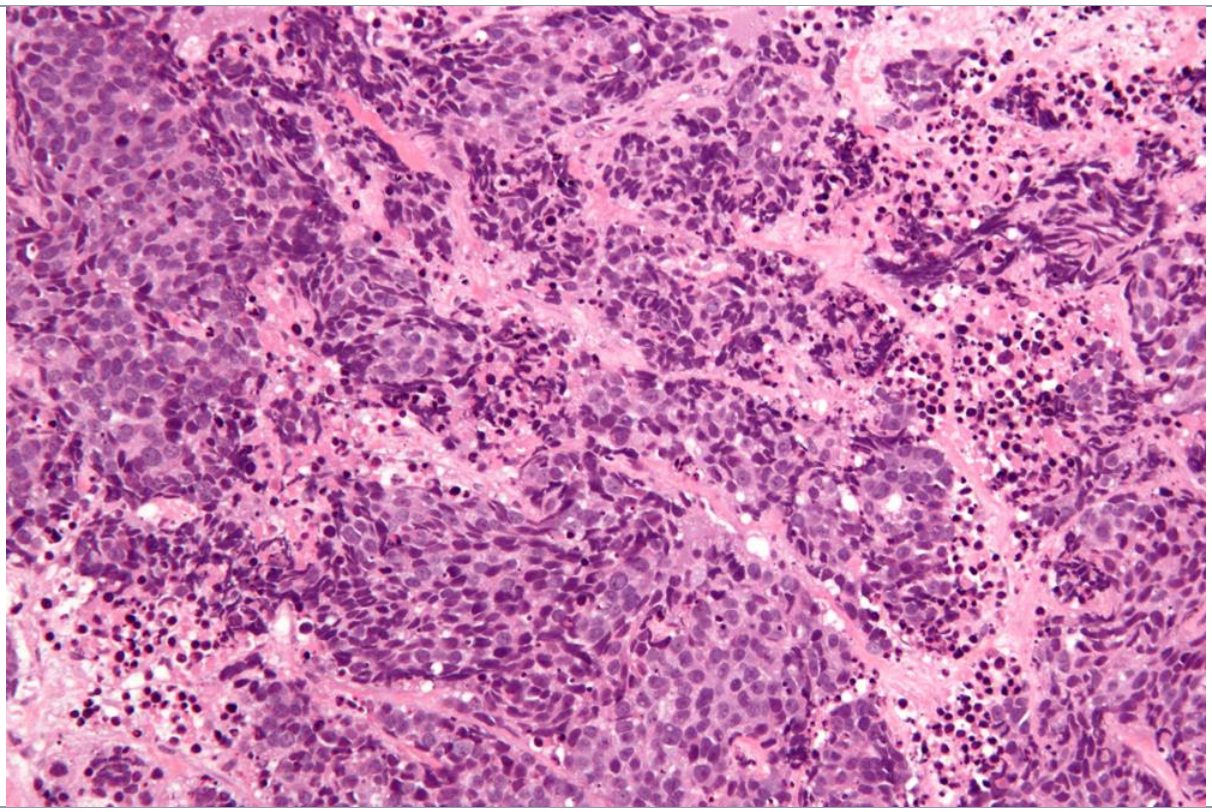


Figure 14. Sinonasal undifferentiated carcinoma (H&E, ×200). Sheets of undifferentiated cells with high mitotic rate and necrosis. Its proximity to the nasal and oral cavities, combined with high tumor cell turnover, generates detectable salivary cfDNA and protein signals. Early detection of this aggressive entity — which has <35% five-year survival at Stage IV — represents the most compelling argument for routine salivary cancer screening in dental practice. Source: Wikimedia Commons.

8. Non-Invasive Devices and Point-of-Care Technologies

Graphene-based electrochemical biosensors functionalized with antibody or aptamer probes achieve picomolar detection limits for salivary cancer markers via voltammetry or electrochemical impedance spectroscopy, readable by portable potentiostats or smartphone attachments within 8–15 minutes [5]. Microfluidic lab-on-chip cartridges integrate sample filtration, nucleic acid extraction, amplification, and optical detection in a single consumable device requiring no laboratory infrastructure.

> **Clinical relevance:** *Chairside deployment timeline: FDA clinical trials for saliva-based OSCC biosensor devices are projected for 2024–2026. If successful, these devices would allow dentists to perform cancer screening as a routine adjunct to the dental examination — transforming every dental visit into a potential life-saving screening event.*

Machine learning algorithms trained on multi-omic salivary datasets generate composite diagnostic scores with AUCs above 0.95 for oral cancer — outperforming any individual biomarker and reducing false positives by approximately 34% compared to single-marker approaches [4,5].

9. Regulatory, Ethical, and Implementation Challenges

No widely approved salivary cancer diagnostic currently exists in clinical practice. The regulatory pathway requires demonstration of analytical validity (precision, accuracy, interference resistance), clinical validity (sensitivity and specificity in the intended population), and clinical utility (evidence that the test improves patient outcomes) — a three-tier evidence standard that current studies satisfy only partially.

Standardization is the most urgent scientific barrier. Without international consensus on collection protocols, processing methods, and analytical platforms, results from different laboratories cannot be meaningfully compared — preventing the meta-analyses needed for regulatory submissions. Ethical considerations include genomic privacy (salivary DNA may reveal germline variants beyond the targeted cancer markers) and management of incidental findings.

Meta-analytic evidence confirms that heterogeneity is not merely a statistical inconvenience but a genuine translational barrier. Wijaya et al. (2026) found QUADAS-2 flagged moderate risk of bias in the majority of included studies, and meta-regression failed to identify any single variable — including assay type, collection method, or sample size — as explaining the between-study variance [12]. Owecki and Nijakowski (2025) similarly noted that variable assay platforms across LDH and Chemerin studies prevented direct cross-laboratory comparison despite large pooled effect sizes [14]. Until unstimulated whole-saliva collection (morning, fasting, standardised volume) is adopted universally and validated ELISA or PCR platforms are agreed upon, regulatory approval will remain out of reach. The international research community's call for PROSPERO-registered prospective multicenter trials with transparent reporting is now supported by multiple independent systematic reviews [12–16].

10. Gaps and Future Directions

- Large-scale multicenter validation: Most evidence derives from single-institution cohorts of 10–100 subjects. Prospective trials enrolling thousands of patients with standardized pre-analytical protocols are required for regulatory submission.
- Integrated multi-omic panels: Machine learning-guided optimization of panels combining DNA, RNA, protein, and metabolite data — trained on large, ethnically diverse datasets — will maximize diagnostic performance.
- Point-of-care maturation: Biosensor prototypes must progress through manufacturing scale-up and regulatory approval, targeting >90% sensitivity, >90% specificity, <30-minute time-to-result, and <$50 per test.
- Dental education integration: Oral health professionals represent the most logical first adopters of chairside salivary diagnostics. Training curricula should incorporate salivary diagnostics as a core competency alongside clinical examination skills.
- Egyptian and regional research gap: No systematic review or meta-analysis of salivary biomarkers in oral cancer has been published by Egyptian or regional institutions. The only Egyptian contribution identified is a 2020 narrative review (Ghalwash, British University Egypt) without pooled statistics. Preliminary Egyptian cohort data — including Ghallab et al. (Cairo University, 2017), who reported 100% sensitivity and specificity for IL-6, IL-8, and sCD44 in small OSCC subgroups — remain unvalidated and unpooled. Prioritising prospective, PROSPERO-registered, multicenter Egyptian studies using standardised saliva collection and validated ELISA platforms would directly address this gap and position Egyptian researchers to contribute to the global evidence base.
- Prognostic biomarker development: Current research has focused predominantly on diagnostic accuracy (sensitivity and specificity at a single time-point). Emerging evidence now identifies salivary biomarkers with prognostic value: miR-423-5p predicts shorter disease-free survival; AKR1B10 above 646 pg/mL predicts worse overall survival; 3-methylhistidine correlates with reduced OS; and miR-1307-5p is associated with nodal metastasis and chemoresistance. Future research should systematically

evaluate these prognostic signals in prospective longitudinal cohorts, enabling salivary diagnostics to inform not just early detection but treatment stratification and surveillance.

11. Conclusion

Saliva is not merely a digestive secretion. It is a diagnostically rich liquid biopsy that encodes the molecular events of local and systemic malignancy — encoding them in proteins, nucleic acids, and metabolites that are measurable, reproducible, and increasingly analyzable at the point of care. The immunohistochemical evidence reviewed here confirms that the salivary signal is grounded in real tissue biology: p53 accumulation and Ki-67 dysregulation in oral premalignant and malignant tissues directly generate the cfDNA mutations and mRNA elevations detected in saliva. Multi-marker salivary panels achieve 88–95% sensitivity for oral cancer — performance sufficient for primary screening. The dental operatory, with its regular patient contact and proximity to the oral cavity, is the optimal deployment setting for these technologies. The barrier is no longer scientific proof of concept. It is standardization, scale, and clinical will.

12. References

All references follow Vancouver citation style. In-text citations use superscript numbers [1–16].

1. Munsif A, Nowaczyk A, Fijałkowski Ł, Riaz S, Jamil A. Salivary biomarkers in cancer detection and management. *Acta Biochim Pol.* 2025;72:14410. https://doi.org/10.3389/abp.2025.14410
2. Xiao H, Zhang L, Zhou H, Lee JM, Garon EB, Wong DT. Proteomic analysis of human saliva from lung cancer patients using two-dimensional difference gel electrophoresis and mass spectrometry. *Mol Cell Proteomics.* 2012;11:M111.012112. https://doi.org/10.1074/mcp.M111.012112
3. Koopaie M, Jomehpoor M, Manifar S, Mousavi R, Kolahdooz S. Evaluation of salivary KCNJ3 mRNA levels in breast cancer: a case-control study and in silico analysis. *Open Dent J.* 2022;16:e187421062208100. https://doi.org/10.2174/18742106-v16-e2208100
4. Banavar G, Ogundijo O, Toma R, et al. The salivary metatranscriptome as an accurate diagnostic indicator of oral cancer. *NPJ Genom Med.* 2021;6:105. https://doi.org/10.1038/s41525-021-00257-x
5. Mahalakshmi D, Nandhini J, Meenaloshini G, et al. Graphene nanomaterial-based electrochemical biosensors for salivary biomarker detection: a translational approach to oral cancer diagnostics. *Nano Transl Med.* 2025;4:100073. https://doi.org/10.1016/j.ntm.2025.100073
6. Bel'skaya LV, Dyachenko EI. Salivary biomarkers in breast cancer: from salivaomics to salivaoncoomics. *Front Biosci Landmark Ed.* 2024;29(7):253. https://doi.org/10.31083/j.fbl2907253
7. Park NJ, Zhou H, Elashoff D, et al. Salivary microRNA: discovery, characterization, and clinical utility for oral cancer detection. *Clin Cancer Res.* 2009;15(17):5473–77. https://doi.org/10.1158/1078-0432.CCR-09-0103
8. Hu S, Arellano M, Boontheung P, et al. Salivary proteomics for oral cancer biomarker discovery. *Clin Cancer Res.* 2008;14(19):6246–52. https://doi.org/10.1158/1078-0432.CCR-07-5037
9. Wong DT. Salivary diagnostics powered by nanotechnologies, proteomics and genomics. *J Am Dent Assoc.* 2006;137(3):313–21. https://doi.org/10.14219/jada.archive.2006.0206
10. Zhang L, Farrell JJ, Zhou H, et al. Salivary transcriptomic biomarkers for detection of resectable pancreatic cancer. *Gastroenterology.* 2010;138(3):949–57. https://doi.org/10.1053/j.gastro.2009.11.010
11. Humayun S, Ram Prasad V. Expression of p53 protein and ki-67 antigen in oral premalignant lesions and oral squamous cell carcinomas: an immunohistochemical study. *Natl J Maxillofac Surg.* 2011;2(1):38–46. https://doi.org/10.4103/0975-5950.85852
12. Wijaya R, Nakashima T, Hirai T, et al. Diagnostic accuracy of salivary biomarkers for oral squamous cell carcinoma: a systematic review and meta-analysis. *Cancers.* 2026;18(2):456. https://doi.org/10.3390/cancers18020456

13. Shaukat A, Al-Hussaini A, Hamdan M, et al. Diagnostic accuracy of salivary MMP-9 for early detection of oral squamous cell carcinoma: a systematic review and meta-analysis. *BMC Oral Health*. 2026;26(1):112. https://doi.org/10.1186/s12903-026-05112-3

14. Owecki M, Nijakowski K. Salivary LDH, MMP-9 and Chemerin as potential biomarkers of oral cancer: a systematic review and meta-analysis. *Int J Mol Sci.* 2025;26(4):1823. https://doi.org/10.3390/ijms26041823

15. Hashmi AA, Naqvi SQH, Edhi MM, et al. Salivary microRNAs as biomarkers for oral squamous cell carcinoma: a systematic review. *Int J Oral Maxillofac Surg.* 2026;55(3):301–310. https://doi.org/10.1016/j.ijom.2025.10.012

16. Antonelli A, Guerrini L, Pini A, et al. Salivary metabolomics for oral cancer detection: a systematic review. *Oral Oncol Rep.* 2024;10:100183. https://doi.org/10.1016/j.oor.2024.100183